

Consider the above balanced equation. If the decomposition of $\text{C}_4\text{H}_8\text{N}_8\text{O}_8(\text{s})$ produces a total pressure of 3.7 atm, what is the partial pressure of elemental gas?

- ☐ A. 0.67 atm [16%]
☐ B. 1.3 atm [22%]
☒ C. 2.0 atm [43%]
☐ D. 2.2 atm [18%]

Correct

43% answered correctly

Explanation:

Partial pressure of elemental gas formed from a decomposition reaction

$\text{C}_4\text{H}_8\text{N}_8\text{O}_8(\text{s})$

$3 \text{CO}_2(\text{g}) + \text{C}(\text{s}) + 2 \text{H}_2\text{O}(\text{g}) + 2 \text{H}_2(\text{g}) + 4 \text{N}_2(\text{g})$

Dalton's Law of Partial Pressures

$X_{\text{elemental gas}} = \frac{6 \text{ mol}}{11 \text{ mol total}} = 0.54$

$P_{\text{elemental gas}} = X_{\text{elemental gas}} \times P_{\text{total}} = 2.0 \text{ atm}$

Dalton's law of partial pressures states that for a mixture of gases, the **total pressure** P_{total} is equal to the sum of the partial pressures of each constituent gas at constant volume and temperature.

$$P_{\text{total}} = \sum P_i = P_1 + P_2 + P_3 + \dots + P_i$$

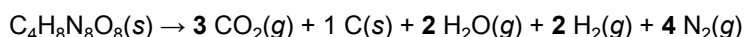
The amount of each gas present in the mixture can be expressed as a **mole fraction** (X_i) (ie, the moles of gas component n_i divided by the total number of moles n_{total}).

$$X_i = \frac{n_i}{n_{\text{total}}} = \frac{n_i}{n_1 + n_2 + n_3 + \dots + n_i}$$

Multiplying X_i by P_{total} yields the partial pressure of the gas component P_i in the mixture.

$$P_i = X_i \cdot P_{\text{total}} = \frac{n_i}{n_{\text{total}}} \cdot P_{\text{Total}}$$

In this experiment, a sample of $\text{C}_4\text{H}_8\text{N}_8\text{O}_8(\text{s})$ is thermally decomposed, generating enough gas to exert a pressure of 3.7 atm. According to the **stoichiometric coefficients** obtained from the **balanced chemical equation**



For every 1 mole of $\text{C}_4\text{H}_8\text{N}_8\text{O}_8(\text{s})$ that decomposes, **11 total moles of gas** are produced. Of the moles of gas produced, **6 moles are elemental gases** (2 mol $\text{H}_2(\text{g})$ and 4 mol $\text{N}_2(\text{g})$), which is 54% of the total gas mixture, as indicated by the mole fraction of elemental gas:

$$X_{\text{elemental gas}} = \frac{n_{\text{H}_2} + n_{\text{N}_2}}{n_{\text{total}}} = \frac{6 \text{ mol}}{11 \text{ mol}} = 0.54$$

The partial pressure of elemental gas can be calculated by multiplying the mole fraction of elemental gas by the total pressure:

$$P_{\text{elemental gas}} = X_{\text{elemental gas}} \cdot P_{\text{total}} = (0.54) \cdot (3.7 \text{ atm}) = 2.0 \text{ atm}$$

Therefore, the elemental gases $\text{H}_2(\text{g})$ and $\text{N}_2(\text{g})$ account for 2.0 atm of the total pressure exerted by the gas mixture.

(Choice A) 0.67 atm is the partial pressure of *only* $\text{H}_2(\text{g})$ in the gas mixture.

(Choice B) 1.3 atm is the partial pressure of *only* $\text{N}_2(\text{g})$ in the gas mixture.

(Choice D) A partial pressure of 2.2 atm is obtained if $\text{C}(\text{s})$ is erroneously included in the moles of elemental gas, which would make $X_{\text{elemental gas}} = \frac{n_{\text{H}_2} + n_{\text{N}_2} + n_{\text{C}}}{n_{\text{total}}} = \frac{7 \text{ mol}}{12 \text{ mol}}$.

Educational objective:

The total pressure exerted by a mixture of gases is equal to the sum of the partial pressures of each gas component.
The partial pressure of each gas component is proportional to its mole fraction.

Time spent: 0 sec

Subject:
General Chemistry

QID: 403331

System:
Thermodynamics, Kinetics & Gas Laws